ELECTRIC VEHICLE MANAGEMMENT SYSTEM USING ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

INDIVIDUAL THESIS

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Individual component: Adaptive Battery Health Prediction and Personalized Recommendations

Group ID: R26-IT-029

Report submitted in partial fulfilment of the requirements for the

BSc (Hons) degree in Information Technology Specializing in Information Technology

Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

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ABSTRACT

As Electric Vehicles (EVs) become more prevalent across the board, the need for smart battery management systems to enhance battery performance, reliability, and longevity has grown. The lithium-ion battery is one of the most costly and crucial parts of an EV, and its deterioration has a significant impact on driving range, efficiency and maintenance expenses. There are various factors that impact the battery degradation such as number of charging cycles, operating temperature, battery age and usage condition. But conventional Battery Management Systems (BMS) mainly monitor existing battery states including State of Charge (SOC) and provide little detail about battery health and battery maintenance in the future.

This research proposes an Adaptive Battery Health Prediction and Personalized Recommendation System as a single component of an integrated Electric Vehicle Management System (AEVS) using Artificial Intelligence and Machine Learning. The intended component is designed to forecast the future state of battery health (SoH) and give individual recommendations to facilitate proactive battery management. Historical battery degradation data was leveraged to implement and evaluate two supervised machine learning algorithms. Two supervised machine learning algorithms, Random Forest and Extreme Gradient Boosting (XGBoost) were implemented and evaluated. The prediction framework focuses on three prediction time horizons: 3 months, 6 months and 12 months battery health. The predictive models were developed and evaluated by performing data preprocessing, feature selection, model training and model evaluation. The performance of the models was assessed through performing Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Coefficient of Determination (R^2).

The experimental results showed that the forecasting results of Random Forest were better than that of XGBoost for the three forecasting periods. Furthermore, a rule-based recommendation mechanism is provided that considers user-provided charging and driving behaviors to provide user-specific battery protection suggestions. The component combines battery health forecasting and behavior-based recommendations, offering a user-friendly decision support tool for grasping the battery's future health and implementing better battery usage practices.

Keywords: Electric Vehicles, Battery State of Health, Battery Degradation Prediction, Random Forest, XGBoost, Personalized Recommendations.

ACKNOWLEDGEMENTS

Many thanks to all the people who guided, supported and encouraged me during the execution of my individual research component 'Adaptive Battery Health Prediction and Personalized Recommendation System' from the research project titled 'Electric Vehicle Management System Using Artificial Intelligence and Machine Learning'.

I would like to express my sincere gratitude to my supervisor Mrs. Uthpala Samarakoon and Co-supervisor Mr. Nelum Amarasena for their constant guidance, useful suggestions, constructive feedback and encouragement during the development and research work. They provided their expertise and support to define the direction of the research and to increase the quality of this study.

Finally, I would like to thank the Sri Lanka Institute of Information Technology (SLIIT) and the Department of Information Technology (IT) of the Faculty of Computing for the academic environment, knowledge and resources that were provided for the successful completion of this research.

In addition, I would like to express my appreciation to all members of my research team for their cooperation, technical support and valuable contributions with the development and integration of the overall Electric Vehicle Management System. I also want to thank the researchers and agencies that put relevant scientific publications, data sets, and technical resources out there for free for use in developing and evaluating the machine learning models that were applied in this research.

Finally I would like to thank my family and friends for their ongoing encouragement, patience and support throughout my educational career.

This research was done as part of my undergraduate academic requirements at SLIIT. This individual research element did not receive any external funding.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
BMS	Battery Management System
DC	Direct Current
EV	Electric Vehicle
LLM	Large Language Model
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MDSS	Mobile Decision Support System
ML	Machine Learning
NCM	Nickel Cobalt Manganese
R^2	Coefficient of Determination (R-squared)
RAG	Retrieval-Augmented Generation
RF	Random Forest
RMSE	Root Mean Squared Error
SOC	State of Charge
SOH	State of Health
SVM	Support Vector Machine
XGBoost	Extreme Gradient Boosting

1 INTRODUCTION

1.1 Overall project context

Electric Vehicles (EVs) are gaining significant attention in the field of sustainable mobility due to their potential to lower reliance on fossil fuels and increase energy efficiency [1]. But battery degradation is one of the biggest challenges as it impacts vehicle performance, driving range, maintenance costs, and battery life.

There are many reasons why lithium-ion batteries are used in EVs, among them being high energy density and high efficiency. These batteries do lose their capacity over time due to ageing, charging, temperature extremes and operating conditions [2]. Hence, prediction of battery health is significant to enhance long term battery management.

In response, the overall research project "Electric Vehicle Management System Using Artificial Intelligence and Machine Learning" was designed to create an intelligent EV decision support system that overcomes the challenges. It is composed of four major parts:

I. **AI-Based EV Operational Anomaly Detection:** Detects unusual operating conditions of a vehicle.
II. **Adaptive Battery Health Prediction and Personalized Recommendation:** Performs battery health prediction and offers battery maintenance suggestions.
III. **Intelligent EV Charging Optimization and Range Prediction:** Enables charging decisions, driving range prediction.
IV. **AI-Based EV Virtual Assistant:** Gives intelligent support concerning EV usage and upkeep.

These components work together with an EV management application to deliver predictive data and solutions for vehicle management.

1.2 Individual component context

This is individual research on the topic of "Adaptive Battery Health Prediction and Personalized Recommendation". Its primary use is to forecast future EV battery State of Health (SOH) and offer customized battery maintenance recommendations.

The SOH of a battery measures how well it is doing compared to the initial capacity. While State of Charge (SOC) refers to the charge level of the battery now, SOH represents the battery's state of degradation after extended usage.

SOH is calculated as:

State of Health (SOH) % = (Actual Battery Capacity / Rated Battery Capacity) × 100

To forecast battery health, two supervised machine learning algorithms are employed in this research namely, Random Forest Regression and Extreme Gradient Boosting (XGBoost) Regression in three different forecasting periods: 3 months, 6 months and 12 months.

Historical battery degradation data are used to train and evaluate these models. They make predictions and compare their performance using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 Score.

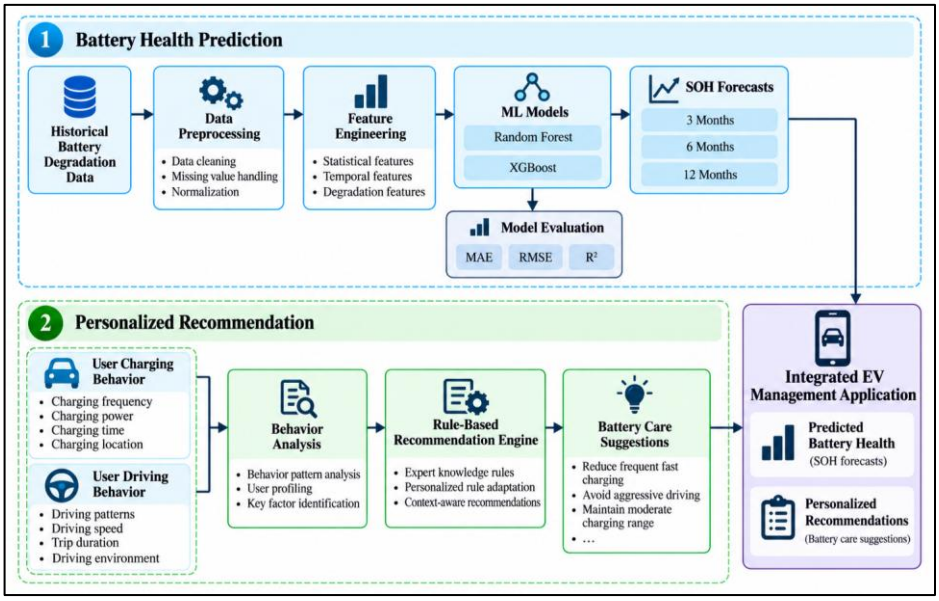


Figure 1 : Overview of the Adaptive Battery Health Prediction and Personalized Recommendation Component

Moreover, a rule-based recommendation engine is used to evaluate the user input charging and driving pattern to recommend battery care practices. The prediction and recommendation processes run independently and the results are then shown in the integrated EV management application as in Figure 1.1.

1.3 Problem statement

The degradation of the battery is one of the most critical issues for EV owners as it affects the battery capacity, driving range and overall vehicle performance over time. The current Battery Management Systems (BMS) can be used to track parameters like voltage, temperature, SOC, but may not be able to track enough information to help understand future battery health [7].

If the owner of an electric vehicle doesn't have predictions in the long term, they can have a hard time planning the maintenance or future condition of the battery. Additionally, simply giving numerical battery health predictions to users may not be enough to help them adopt better charging and driving habits.

Multi-horizon forecasting combined with personalized and practical recommendations is also a critical challenge when applying machine learning techniques to battery health estimation [8].

Thus, this study aims to solve the issue of the EV battery SOH prediction over 3, 6 and 12 months by employing the Random Forest and XGBoost algorithms and to provide individually tailored battery maintenance advice according to users' usage pattern.

1.4 Research gap

There are previous works that have shown the use of machine learning to predict battery degradation from historical battery data [3] and [4]. But there are some drawbacks.

Most existing methods are related to present SOH estimation or battery degradation under experimental conditions. These methods may not be fully representative of the various operating conditions encountered in the actual operation of real EVs [6].

In addition, very little attention is paid to the integration of several forecasting periods in the future and the provision of personalised recommendations in one application that will be used by the user. While driving and charging behavior can impact battery degradation [5], some predictions of degradation do not always become practical battery care recommendations.

To overcome these disadvantages, this research compares the performance of Random Forest and XGBoost in predicting the SOH at 3-, 6-, and 12-month time horizons, and attempts to combine the prediction model with a behavior-based recommendation model.

1.5 Aim and objectives

1.5.1 Main Aim

The main aim of this research is to develop and evaluate a machine learning-based EV battery health prediction system using Random Forest and XGBoost regression models

and provide personalized battery maintenance recommendations based on users' charging and driving habits.

1.5.2 Specific Objectives

I. **Data Preparation:** Preprocess and analyze battery degradation data for building a machine learning model.
II. **Model Development:** Develop Random Forest and XGBoost regression models to predict battery SOH for 3, 6, and 12 months.
III. **Model Evaluation:** Compare the accuracy of prediction of the models by using MAE, RMSE and R2 to find the best model.
IV. **Personalized Recommendations:** Create a mechanism that recommends battery care based on rule sets and battery charging/driving history.
V. **System Integration:** Integrate battery health prediction and recommendation into the overall EV management application.
VI. **System Evaluation:** Evaluate model performance, model prediction consistency, and limitations of the model developed component.

1.6 Research questions or hypotheses

The main goal of this research is to evaluate the ability of machine learning techniques to forecast the future of EV battery State of Health (SOH) and to suggest personalized recommendations for battery maintenance. The study mainly focuses on comparing Random Forest and XGBoost regression models for multi-horizon battery health prediction.

The main question of research is:

What is the performance of the Random Forest and XGBoost regression models in predicting future EV battery State of Health after 3, 6 and 12 months and how can they be used to inform personalized recommendations for Battery Management?

In answering this question, the following research questions are considered:

I. RQ1: How to process and utilize historical battery degradation data for battery health prediction?
II. RQ2: What is the ability of Random Forest and XGBoost to predict the SOH of batteries after 3, 6 and 12 months of usage?

III. RQ3: What is the prediction performance (in terms of MAE, RMSE and R^2) of the machine learning model?
IV. RQ4: How to derive personalized battery maintenance recommendations from charging and driving behavior?
V. RQ5: What is the possible fusion of prediction and recommendation mechanisms for integration in an EV management application?

The hypothesis of the research is that Random Forest and XGBoost can be used to find relationships within historical battery degradation data that can be used to predict the battery's future SOH. Their accuracy in predicting however, may differ from one period of forecasting to another and from one set of characteristics of the data to another.

1.7 Scope limitations and assumptions

This study concentrates on machine learning approaches to build an EV battery health prediction and personalized recommendation module. The study employs the historical battery degradation data to train and tests the regression models: Random Forest and XGBoost for predicting future battery SOH for a period of 3, 6 and 12 months.

Data preprocessing, feature selection, model development, performance evaluation, and integration with the entire EV management application are all covered in the research. The performance of the model is judged by Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 Score. Further, a rule-based recommendation mechanism gives suggestions for battery maintenance based on the battery maintenance data and the user's battery charging and driving behavior.

There are some restrictions, however, in the study. The prediction models are built based on a public battery degradation dataset, which might not be comprehensive enough to cover all EV models, battery types and real-world driving conditions. In addition, the data set lacks all of the behavioral data necessary for personalized recommendations. Thus, the prediction and recommendation processes are made independently.

The research assumes that there exist useful patterns in the historical battery degradation data that can be used to predict the battery health in the future, as well as that the selected battery degradation data set is appropriate for the evaluation of both regression models. It also assumes that users enter the correct charging and driving data when using the personalized suggestions.

The predictions are not intended to give a precise battery life or replacement date but to assist battery management decision making.

1.8 Individual contribution

The Adaptive Battery Health Prediction and Personalized Recommendation component is a part of the overall Electric Vehicle Management System Using Artificial Intelligence and Machine Learning and is being developed by this individual research.

The primary contribution is the development and comparison of the Random Forest and XGBoost regression models to predict battery SOH after 3, 6 and 12 months. This involves creating historical battery degradation data, choosing features of interest, building predictive models, and assessing the models' accuracy with respect to MAE, RMSE, and R^2.

Random Forest has lower prediction errors than XGBoost in the forecasting periods considered based on experimental results. So, Random Forest was chosen as the model of interest because of its capabilities when it comes to battery health prediction.

Another contribution is an analysis of the user-provided charging and driving data that is based on a rule-based approach to suggesting maintenance of the battery. Lastly, the prediction feature and the recommendation feature are embedded into the complete EV management application, and users can receive future battery health projections and the practical battery care advice within a single application.

1.9 Thesis structure

This thesis is composed of nine chapters, which present the research, development, implementation and evaluation of the Adaptive Battery Health Prediction and Personalized Recommendation component.

Chapter 1 - Introduction: This chapter provides an overall background of the project, individual component, problem statement, research gap, project objectives, research questions, scope and individual contribution.

Chapter 2 - Literature Review and Related Work: A review of the existing literature on EV battery degradation, State of Health (SOH) prediction, machine learning techniques, and battery management systems. It also points to the lack of good approaches.

Chapter 3 - Research Methodology: Describes the Research Design, Data Set Preparation, Experimental Procedures, Development of the models Random Forest and XGBoost, Evaluation Metrics and Data Analysis Methods.

Chapter 4 - Product and Solution Design: It details the system requirements, the architecture of the components, the decisions that had to be made in the design, and the EV management system and its relationship with the battery health prediction component.

Chapter 5 - Implementation, integration and testing: This chapter covers technologies utilized, implementation of prediction models and recommendation mechanisms, system integration, software testing and deployment.

Chapter 6 - Results and Validation: Experimental results, performance comparison of models, technical testing outcomes, and validation of developed component.

Chapter 7 - Discussion: Interprets the research findings, compares them with the existing research, discusses the research contributions, implications and limitations.

Chapter 8 - Commercialization and Sustainability: Discusses the possible applications, target users, commercial interest, scalability and sustainability of the proposed solution.

Conclusion and Future Directions: Summarizes the research findings, achievement of objectives, individual contributions and possible improvements for future research. The references, glossary, and appendices with supporting research materials and additional documentation follow the thesis

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This chapter summarizes the previous work on EV battery degradation, State of Health (SOH) forecasting, machine learning methods, and battery management for individual batteries. The primary goal is to gain insights into current practices, their pros and cons, and the research gaps where the proposed Adaptive Battery Health Prediction and Personalized Recommendation component would fit in.

A focused narrative literature review was conducted from the selected peer-reviewed journal articles, conference publications and technical documentation. The review includes key studies that were published primarily from 2019 to 2025 alongside some of the seminal works on Random Forest and XGBoost. The studies chosen are those dealing with battery degradation, battery SOH estimation, battery lifetime prediction and real-world battery monitoring.

Studies were considered in terms of prediction methods, character of the data, applied model performance, evaluation metrics and practical applications. Random Forest and XGBoost are two regression algorithms that are implemented and evaluated for this research and so these were given special attention.

There was also a review of the existing EV battery monitoring products to get an understanding of how information on battery health is presented to the user. The results serve as a basis for the proposed multi-horizon SOH Prediction and Personalized Recommendation Framework.

2.2 Domain and theoretical foundations

Because of their relatively long service life, charging efficiency and high energy density, lithium-ion batteries are widely used in electric vehicles. These batteries, however, suffer from slow degradation due to electrochemical reactions, charging/discharging cycles, temperature changes, and operating conditions [2].

The degradation of batteries primarily takes place through calendar aging and cycle aging. Calendar aging is a process of battery deterioration with the passage of time and

cycle aging is due to repeated charging and discharging. These processes can lead to losses of capacity and internal resistance which will impact battery performance [2].

Battery State of Health (SOH) is an important indicator used to evaluate the condition of a battery compared with its original condition. For this research SOH is represented by battery capacity:

SoH(%) = Current Battery Capacity/Rated Battery Capacity × 100

For instance, a battery that has 90% SOH will have a capacity of about 90% of the rated capacity. Similarly, as SOH reduces, so may the energy available to drive.

The State of Charge (SOC) and SOH are two different characteristics of the battery. SOC is the charge of a battery at a specific time, and SOH is the condition of a battery over time. The traditional Battery Management System (BMS) monitors battery voltage, current, temperature, and SOC, which are crucial for ensuring safe and efficient battery operation [7].

An alternative way of estimating battery health is to use machine learning and historic degradation data. Regresional models can be used to reveal possible non-linear relationships between battery attributes and SOH without the need to build a full electrochemical model.

This study is particularly relevant with respect to two machines learning algorithms.

Random Forest Regression: Random Forest is an ensemble learning model that utilizes predictions made by several decision trees. A tree is trained on sampled data and the final regression prediction is made by taking the average of the tree outputs. This method can be used to detect non-linear relationships and to minimize the prediction variance [9].

XGBoost Regression is an extreme gradient boosting (XGBoost) ensemble learning algorithm. Every new tree is pruned to minimize the number of misclassifications made by the old trees. XGBoost also has regularization techniques to regulate the complexity of the model and boost generalization [10].

Both algorithms can be applied to structured numerical data sets and performance metrics of regression can be used to assess them.

Mean Absolute Error (MAE) is the average of the absolute value of the difference between predicted SOH value and actual value. Generally, the Root Mean Squared Error (RMSE) is more sensitive to larger prediction errors, while the coefficient of determination (R^2) indicates the fit between the regression model and the variations in the target values.

The proposed battery health prediction models are developed and evaluated, based on these theoretical foundations.

2.3 Related research

Battery health prediction using machine learning has garnered a lot of research interest. The previous research work has been done on statistical methods, traditional machine learning algorithms, deep learning techniques and hybrid methods for estimating battery degradation and battery SOH.

In [3] a data-driven approach to estimate lithium-ion battery cycle life from initial charging and discharging data was developed. They found machine learning algorithms could predict useful degradation patterns before substantial battery degradation had taken place. The study was primarily aimed at predicting battery cycle life but not forecasting SOH for a specific number of months.

The authors in Roman et al. [4] introduced a machine learning pipeline for battery SOH estimation based on charging measurements. They implemented feature engineering, feature selection, and regression modeling in their study to estimate battery condition. Data preparation and predictive uncertainty were shown to be important aspects of the research. It mainly addressed SOH estimation, however, it didn't integrate battery maintenance recommendations.

Lu et al. [11] proposed a deep learning-based framework for SOH estimation using domain adaptation techniques. This study employed commercial battery data and showed that no further degradation testing needs to be performed on each target battery to estimate its SOH. The method, however, is more complex for modeling techniques and only involves battery health estimation, not a battery forecasting and recommendation system.

Jiang et al. [5] explored battery health monitoring using extreme learning machines and feature selection methods. The study considered practical accessibility of the battery health indicators with real world driving. Its findings highlighted the necessity to consider the balance between prediction accuracy and feature accessibility. The study did not particularly study an integrated 3, 6 and 12 months SOH forecasting and recommending mechanism, however.

Liu et al. [6] presented a multi-modal deep learning framework based on operational data of 300 EVs for three years. Their study pointed out the significance of real-world EV data and the discrepancies between laboratory conditions and real-world ones. The study did not specifically focus on the multi-horizon forecasting and rule-based recommendation approach shown to be effective in the research, but rather on the approach to forecasting and recommending based on the data-driven model of SOH evaluation.

Pandit and Ahlawat [12] used NASA battery degradation datasets to compare Random Forest, XGBoost and Support Vector Machine (SVM) algorithms. They found that XGBoost had the best overall performance in their experiments and used a consistent assessment framework. The paper did, however, report that model performance varies across battery types, suggesting that the choice of algorithm is dependent on the characteristics of the data sets and evaluation conditions.

The discussed studies show various methods of battery health estimation and prediction. To determine their applicability to this research, these studies were summarized and key contributions and limitations were listed as shown in Table 2.1. This comparison justifies the choice of Random Forest and XGBoost and emphasizes the need of comparing them in the same forecasting context.

Table 1:Comparison of Related Battery Health Prediction Research

Study	**Method**	**Main contribution**	**Limitation relative to this research**
Severson et al. (2019) [3]	Machine learning	Early battery cycle-life prediction	Does not focus on monthly SOH forecasting

Roman et al. (2021) [4]	ML regression pipeline	SOH estimation using engineered features	No integrated behavioral recommendation mechanism
Lu et al. (2023) [11]	Deep learning	SOH estimation using domain adaptation	More complex approach with a different prediction objective
Jiang et al. (2024) [5]	Extreme learning machine	Practical feature-based battery health monitoring	Does not specifically address multi-horizon prediction and recommendations
Liu et al. (2025) [6]	Multi-modal deep learning	SOH estimation using real-world EV data	Different forecasting and application objectives
Pandit and Ahlawat (2025) [12]	RF, XGBoost, SVM	Comparative model evaluation	Does not integrate the proposed behavioral recommendation workflow

As indicated in Table 2.1, it has been found that machine learning is capable of accurately analysing the battery degradation pattern. The studies vary in their prediction objectives, training sets and evaluation methods, however. So, it is not always possible to compare the reported prediction accuracy directly.

The current research extends these findings by using the same evaluation measures to assess the Random Forest and XGBoost for the three SOH forecasting periods. It also brings an independently developed rule-based recommendation mechanism for enhancement of practical application of the battery health prediction.

2.4 Existing systems products and techniques

There are several commercial systems and technical solutions that can be used to support EV battery monitoring, health estimation and maintenance planning. They are used to give vital data on battery state, energy use, charging and vehicle performance.

The traditional Battery Management System (BMS) is embedded in Electric Vehicles (EVs) to keep track of key battery parameters including voltage, temperature, current and SOC. They serve primarily as protection, balancing, thermal control and charging

control for the battery. Some advanced BMSs can also offer SOH estimation and diagnostic features [7]. The long term forecasting and the user specific recommendations are different for each system, however.

Geotab EV Performance is an electric vehicle fleet monitoring commercial solution. It offers information on battery health, battery capacity usage and driving range analysis. The system allows fleet operators to monitor the performance of the batteries over time and can help with vehicle maintenance and fleet planning [13].

Another EV monitoring platform, Recurrent Connect offers insights into battery health, driving range, charging, and performance. It also provides details on potential battery usage effects due to charging and driving behavior. These features aid EV owners in understanding battery condition, and in making decisions on battery maintenance [14].

Besides commercial products, regression algorithms and deep learning techniques for estimating SOH and analysing degradation are also used in machine learning-based approaches. These techniques can offer prediction capabilities, provided the appropriate datasets, model evaluation and integration with user-facing applications, are used.

While these solutions offer useful features for battery monitoring and guidance, they are not necessarily meant for similar purposes and technical approaches as the proposed research. Table 2.2 provides a comparison of the main functionalities of the existing systems and techniques with the proposed system to highlight its specific functionalities on multi-horizon forecasting and comparative machine learning evaluation.

Table 2 : Comparison of Existing Battery Management Systems and Techniques

System or technique	**Main functionality**	**Relevance or limitation**
Traditional BMS [7]	Battery monitoring, protection, SOC and SOH estimation	Future forecasting and personalized guidance depend on system capabilities

Geotab EV Performance [13]	Battery health and range monitoring for EV fleets	Primarily focused on fleet performance analysis
Recurrent Connect [14]	Battery health, range, charging, and driving insights	Provides user guidance but uses a different monitoring approach
ML-based prediction techniques [8]	Battery SOH estimation and degradation prediction	Require appropriate training data, validation, and application integration
Proposed component	3-, 6-, and 12-month SOH prediction with rule-based recommendations	Combines RF/XGBoost comparison and personalized guidance within the EV management application

It is observed from Table 2.2 that current EV management solutions are already capable of providing diverse EV battery monitoring and decision support functions. But their usefulness is based on their target group, the data provided in the vehicles and the technical implementation.

The research proposed is specifically targeting a certain implementation, which involves comparing two machine learning algorithms for future SOH forecasting and combining the algorithm selected into a separate recommendation mechanism on the same platform. The goal is to not only give technical battery health estimates, but also to give battery health practical advice through the overall EV management application.

2.5 Gap synthesis and conceptual framework

The surveyed literature reveals that there are some issues with the battery SOH prediction, dataset appropriateness and model selection, and practical battery management.

First, there are many studies that are available that concentrate on both current SOH estimation, battery cycle-life prediction or degradation analysis under specific experimental conditions, respectively [3, 4]. These approaches are useful for providing information, but the prediction goals do not always align with the estimation goal for battery SOH for specific future time horizons, e.g., 3, 6 and 12 months.

Secondly, many battery models use the experimental data from the past, which may not accurately reflect real-life EV operating conditions. Degradation behavior may be affected by battery chemistry, ambient temperature, and usage. Thus, the prediction of accuracy under experimental conditions may not necessarily reflect the performance of different commercial EVs [6].

Third, the performance of machine learning models depends on the dataset used, features extracted, training processes used, and the testing methods [12]. Hence for this research, performance comparisons between two algorithms, Random Forest and XGBoost, is required to determine which one is more appropriate for the battery degradation data.

Finally, numbers alone might not offer enough assistance in everyday maintenance decisions. While it is important and useful to have existing products that provide guidance, the integration of battery forecasting with well defined behavior recommendations is still a valid application-level research goal.The proposed research provides a conceptual framework of two distinct yet connected processes to address these challenges.

The first one is dedicated to the battery degradation data for battery health prediction. Data Pre-processing, Relevant features preparation for Machine Learning. Random Forest and XGBoost regression models are trained and evaluated by MAE, RMSE and R^2. Based on the evaluation results, the selected model is applied to predict the future battery SOH for 3, 6 and 12 month time horizons, with the data's temporal information and forecasting assumption.

The second process is to concentrate on the personalized battery maintenance advice. Charging and driving information provided by the user is processed according to a set of behavioral rules and transformed to practical charging advice for the battery. This recommendation mechanism is separate from the machine learning predictive process since all the behavioral attributes described in the historical degradation data set are not sufficient to train a personalized recommendation process.

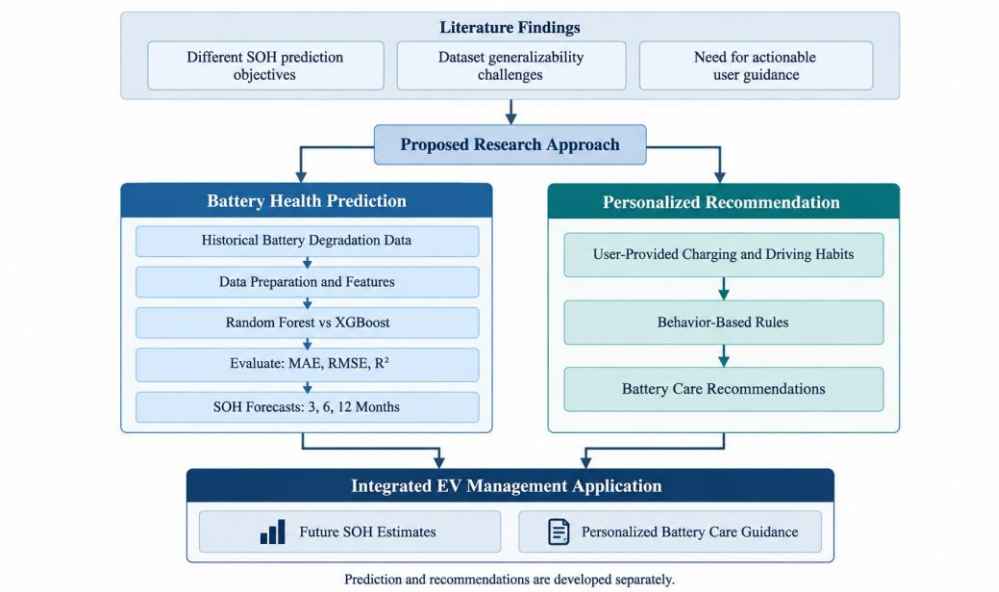


Figure 2 : Gap Synthesis and Conceptual Framework of the Proposed Component

The conceptual framework is presented in figure 2.1, which was developed based on literature gaps. It provides a view on how the prediction and recommendation systems work separately, prior to their input being amalgamated in the integrated EV management application.Figure 2.1 shows the gap synthesis and conceptual framework of the proposed component. Gap synthesis and conceptual framework of the proposed component is shown in figure 2.1.

The conceptual framework highlights the connection between the research gap and the solution proposed. The battery health prediction process returns future SOH estimates and the recommendation process returns behavioral advice. The combination of these two modes of information allows users to access it in one application.

So this research does not focus on developing a new machine learning algorithm. On the contrary, it is the comparative analysis of the prediction capabilities of Random Forest and XGBoost methods for multi-horizon battery SOH prediction, as well as the incorporation of the chosen prediction method with rules-based personalized recommendations for battery care.

3 RESEARCH METHODOLOGY

3.1 Research design

The present research adopts quantitative-experimental approach to predict the State of Health (SOH) of Electric Vehicle (EV) batteries using machine learning. To assess their predictive capabilities, two supervised regression models, Random Forest and XGBoost, are designed and tested. For their predictive abilities, two supervised regression models: Random Forest and XGBoost are developed and compared for prediction of battery health over 3, 6 and 12 months.

The study is further based on a rule-based recommendation system which offers personalized battery maintenance recommendations based on user-supplied charging and driving pattern. Both features are incorporated into the EV management application.

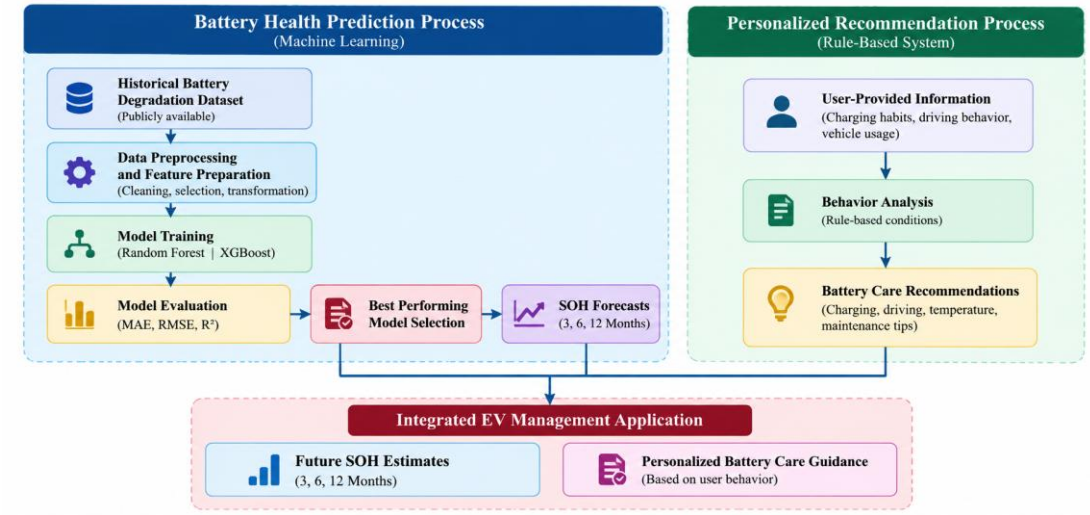


Figure 3 : Research Methodology Workflow of the Proposed Component

The research methodology is represented in Figure 3.1, which includes data preprocessing, training, evaluation, and prediction of SOH using 3, 6 and 12 months of data for both Random Forest and XGBoost models. The user-provided charging and driving behaviors are broken separately, and battery care recommendations are created based on those. These outputs are incorporated into the EV management application.

3.2 Data participants or study materials

In this study, the degradation dataset of lithium-ion batteries used for machine learning is publicly available. The dataset includes records of battery performance measurements which can be utilized to investigate battery capacity deterioration and make the best estimate for battery SOH.

The battery parameters and information are extracted and manipulated to form input features and output prediction targets. The prediction models do not need to be trained with the direct involvement of human participants. Information about charging and driving is separated by application users for personalized recommendation generation.

3.3 Experimental or study procedure

The steps of the experimental process are data preprocessing, feature preparation, model training, model evaluation and system integration.

First, the battery dataset is checked and prepared by fixing missing data, invalid observations, and inconsistencies as needed. The features that are relevant are then chosen to predict future battery SOH.

To train and evaluate the regression models, the same experimental procedure is followed: Random Forest and XGBoost models. Three forecasting periods are included in the research: 3 months, 6 months, and one year. The performance of the models is assessed and compared to determine the most appropriate.

A rule-based recommendation mechanism is used to analyze the user-entered drive and charge patterns separately and provide battery care suggestions. Finally, the prediction and recommendation capabilities are integrated with the EV management application.

3.4 Measures and evaluation metrics

Three classic regression metrics are used for performance evaluation of the Random Forest and XGBoost methods:

- **The Mean Absolute Error (MAE):** is the average absolute difference between the actual and predicted SOH values.

- **Root Mean Squared Error (RMSE):** Calculates prediction errors but assigns more weight to larger errors.
- **Coefficient of Determination (R^2):** represents the proportion of the battery SOH values variation that is accounted for by the model.

A lower MAE and RMSE values suggest a greater accuracy of prediction and a correlation coefficient closer to 1 suggests a better model fit. These metrics have been used to compare the two algorithms over the forecasting periods.

3.5 Data analysis

The battery degradation data is used for analysis to find patterns and relationships that impact battery SOH. To develop machine learning models, data preprocessing and feature preparation are done.

The MAE, RMSE, and R^2 values for each forecasting period are used to compare the prediction results of Random Forest and XGBoost. The results of the experiments show that the model with XGBoost is more accurate in predicting errors than the model with Random Forest in the models evaluated.

The results are also analyzed to gain insight into the consistency of the forecasts for various forecasting periods and understand the restrictions associated with the characteristics of the datasets and the performance of the models.

3.6 Ethics privacy security and responsible AI

The battery degradation data considered in this research is publicly available, mitigating concerns with collecting sensitive personal information for model training.

Charging/driving information provided by the user is treated separately for customizations. This information must be processed securely, only the data that is necessary should be collected and only authorized application functions should be able to access it.

Predictions using machine learning models are designed to assist in making informed decisions about battery management, not to predict battery performance. Considerations for responsible AI include data quality, reliability of prediction, transparency, and clearly communicating model limitations.

3.7 Reproducibility and research artefacts

Reproducibility is assured by providing information about the source of data, the preprocessing steps, the features selected, the model configurations and the evaluation methods.

The primary research artifacts consist of the dataset of batteries prepared for the research, the Python machine learning scripts, the trained Random Forest and XGBoost models, the results of the performance evaluation and the implementation of the rule-based battery recommendation.

The development and evaluation procedures are well documented for future use to compare with other models, improve the system, and extend research into EV battery health prediction.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The Adaptive Battery Health Prediction and Personalized Recommendation component aims to provide Electric Vehicle (EV) users with insights on the future condition of their battery and with practical tips for battery maintenance. The system requirement is determined according to the research goals, users' needs, and integration needs of the entire EV management application.

Requirements are broken down into functional and non-functional. The functional requirements are the primary functions of the component and the non-functional requirements are the characteristics and performance of the system.

The key functions needed for battery health prediction and personalization recommendations are summarized in Table 4.1. These requirements direct the development, integration and testing of the component.

Table 3 : Functional Requirements of the Proposed Component

ID	**Functional Requirement**
FR1	Process historical battery degradation data for machine learning model development.
FR2	Predict battery State of Health (SOH) for 3, 6, and 12 months.
FR3	Evaluate and compare Random Forest and XGBoost models using MAE, RMSE, and R^2.
FR4	Accept user-provided charging and driving behavior information.
FR5	Generate personalized battery maintenance recommendations using predefined rules.
FR6	Display battery health predictions and recommendations through the EV management application.

Non-functional requirements describe quality criteria required to make the suggested component effective. These can be performance, accuracy, usability, security,

reliability and maintainability. The key non-functional requirements considered in system design are summarized in Table 4.2.

Table 4 : Non-Functional Requirements of the Proposed Component

Requirement	**Description**
Performance	Generate battery health predictions within a reasonable response time.
Accuracy	Provide reliable SOH estimates based on model evaluation results.
Usability	Present predictions and recommendations through a clear, user-friendly interface.
Security	Protect user-provided information from unauthorized access.
Reliability	Handle invalid inputs and processing errors appropriately.
Maintainability	Support future model updates and improvements without major changes to the application.

These non-functional requirements support the development of a reliable, secure, and maintainable component suitable for integration into the overall EV management system.

4.2 Component architecture within the complete solution

The proposed Battery Health Prediction component is part of the overall Electric Vehicle Management System Using Artificial Intelligence and Machine Learning. It features operational anomaly detection, battery health prediction and recommendations, intelligent charging optimization and range prediction, and an AI-powered virtual assistant.

The individual component includes two processes that are based on machine learning to predict battery health and on rule-based methods to generate personalized battery recommendations.

Historical battery degradation data are used in the battery health prediction process to preprocess, feature prepare and develop models. Random Forest and XGBoost

regression models are used to assess their performance by selecting more appropriate regression models for predicting future SOH after 3, 6 and 12 months.

The personalized recommendation process works separately on user-specified charging and driving behavior following defined behavioral rules. It provides actionable battery care tips, without needing the inputs of recommendations to be included in the dataset used for training the machine learning model.

The component is integrated into the overall EV management application via integration interfaces to both processes. This architecture will enable the user to view future battery health estimates and customized battery maintenance advice in one app while also facilitating integration with other EV management capabilities.

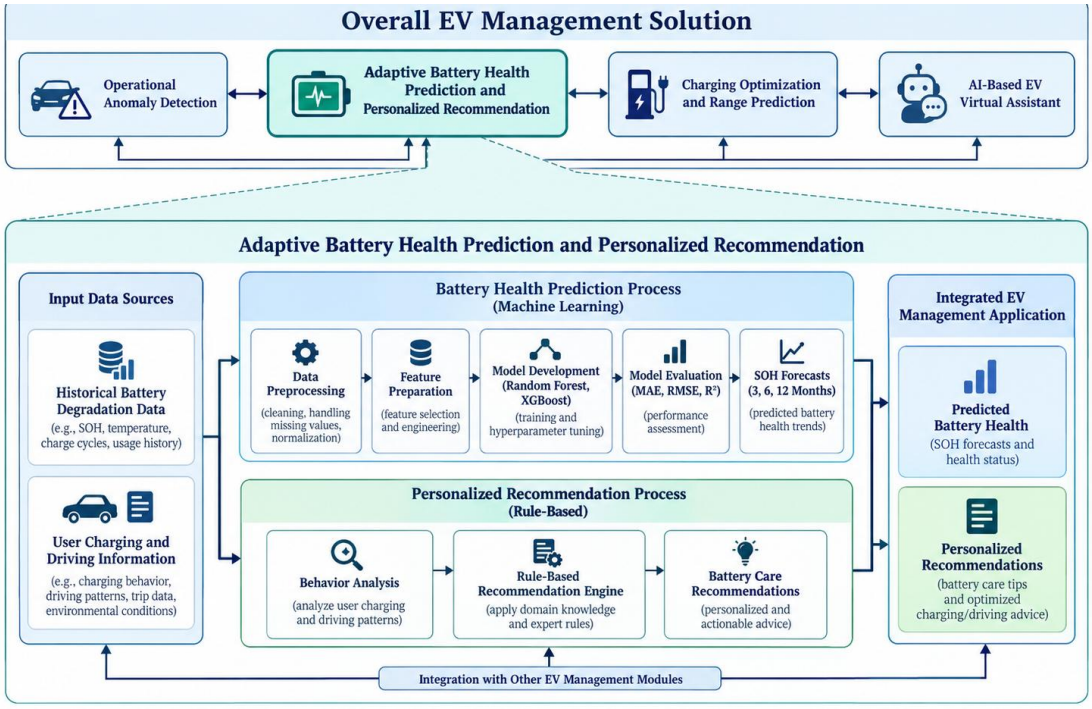


Figure 4 : Component Architecture of the Adaptive Battery Health Prediction and Personalized Recommendation System

The proposed architecture of the component in the whole EV Management System is shown in Figure 4.1. It points out the differences between the prediction and the recommendations processes and demonstrates how the results of these processes are combined when applied.

4.3 Design alternatives and decisions

The design of the battery health prediction component was developed by considering several design alternatives. The major choices involved choosing appropriate prediction algorithms, how to implement the recommendation mechanism, and how to design the component architecture.

Random Forest and XGBoost were chosen as the two algorithms because they can capture nonlinear relationships in structured battery degradation data. The Random Forest approach is based on averaging multiple decision trees, and XGBoost is based on sequential boosting to minimize the prediction error [9] [10].

The same regression metrics were employed for the development and evaluation of both models. Random Forest model outperformed the other models in the component model based on the experimental comparison as it showed minimum errors of prediction during the forecasting periods evaluated.

A rule-based approach was used because it was decided to have a separate machine learning model for personalized recommendations. It was decided that way, because the battery data used for training the personalized recommendation model was not of sufficient quality to capture charging and driving behavior.

A modular architecture was also chosen to maintain the independence of the prediction and recommendation processes. This design makes it easier to maintain and means that individual processes can be changed without having to change the whole application.

4.4 Detailed design

The detailed design is provided which explains how the component processes are used to battery information, generate predictions and deliver personalized recommendations.

Historical lithium-ion battery degradation data is the starting point for the battery health prediction process. The data is analyzed and cleaned (where possible) for missing data, invalid observations, and inconsistencies. Features and SOH prediction targets are then prepared which are relevant to the problem in hand, followed by a process of machine learning.

Random Forest and XGBoost regression models are trained with MAE, RMSE and R2. The selected model is applied to estimate future battery SOH from the dataset, and for forecasting periods of 3 months, 6 months, and 12 months, respectively, according to the temporal representation of the dataset and forecasting assumptions.

Information on charging and driving is given by the user to the recommendation mechanism. The inputs can be charging frequency, fast charging usage, driving distance per day, acceleration and braking patterns. A collection of pre-specified conditions is used to determine if there are any undesirable practices and to provide appropriate battery care suggestions.

The application interface displays the predicted SOH values and personalized recommendations in an easy-to-understand format. The prediction and recommendation processes are kept separated, and the results are merged at the application level.

This design allows for a clear distinction between model-based forecasting, rule-based decision support and user interface functionality.

4.5 Standards and professional practice

The proposed component is developed based on software engineering and responsible AI principles, ensuring reliability, maintainability, security, and usability of the system.

Modularity concepts are used to split the data processing, machine learning prediction, recommendation generation and application integration. This separation makes testing, maintenance and future system improvements much easier.

Data quality is ensured by proper data preprocessing and validation. Performance of the models is assessed based on commonly used regression metrics such as MAE, RMSE and R^2 for clear and informed model selection.

Security and privacy are also factors to consider. Only collect users provided charging and driving information, when necessary, validate it before processing, and secure it by using proper access controls and data handling procedures.

Responsible AI principles are considered by describing predictions as an estimate instead of an actual battery condition. The system also separates the predictions from past battery data and the recommendations based on preprogrammed behavioral rules.

Finally, the developed component is well documented, versioned and tested to ensure its reproducibility and professionalism.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

A machine learning approach with Python was used to develop the Adaptive Battery Health Prediction and Personalized Recommendation component. The implementation of the solution includes data preprocessing, model training, battery health prediction, personalized recommendation generation, and integration with the entire EV management application.

The decision was made to use Python due to its vast library of machine learning tools and its ability to handle data analysis. In this work, the future battery State of Health (SOH) is estimated by using regression algorithms, namely Random Forest and XGBoost. The design consists of a frontend application, backend services, and a machine learning service. It is a modular design and facilitates independent development, testing, and maintenance of individual components.

Table 5.1 summarizes the main technologies relevant to the development of the proposed component. These technologies support the implementation of predictive models and integration with the EV management application.

Table 5 : Technologies Used in Component Development

Technology	**Purpose**
Python	Machine learning development and prediction processing
Pandas and NumPy	Dataset preprocessing and numerical operations
Scikit-learn	Random Forest implementation and model evaluation
XGBoost	Gradient boosting regression model development
Backend services	Communication between the application and prediction component
Frontend application	Display battery health predictions and recommendations
Git and GitHub	Source code management and version control

5.2 Component implementation

The key functions of the component implementation are battery health prediction and personalized recommendations for battery maintenance.

The historical lithium-ion battery degradation data were preprocessed and selected for battery health prediction. The selected battery attributes were used to develop Random Forest and XGBoost regression models.

Three forecast periods (3, 6 and 12 months) were used for training the models to estimate future battery SoH. They were assessed with Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 Score.

Random Forest performed better than XGBoost in terms of prediction errors as per the experimental comparison and was considered as a better prediction model for the forecasting periods considered.

The personalized recommendation mechanism was designed independently based on the set of rules of behavior. It examines the information provided by users related to charging and driving, such as frequency of charging, usage of fast charging, driving distance and driving behavior.

The mechanism also outputs battery care recommendations based on the inputs, such as decreasing fast charging and driving aggressively. The recommendation machine learning training dataset is not involved in the recommendation process. Both functions return values which can be displayed in the built-in EV management application.

5.3 Integration with the other components

The battery health prediction module is part of the overall Electric Vehicle Management System using Artificial Intelligence and Machine Learning.

The full solution also features operational anomaly detection, intelligent charging optimization & range prediction, AI-based virtual assistant, and the battery health prediction module.

The frontend application allows the user to access battery health information as well as to enter the relevant behavior information. The backend acts as a message bus to orchestrate communications with prediction and recommendation services.

The machine learning service takes as input the predictions and outputs the estimated SOH values for the chosen forecasting periods. On its own, the recommendation mechanism processes driving and charging data and generates personalized battery care recommendations.These outputs are integrated into the application to provide forecasts of battery health and maintenance advice in one place.

The modular integration permits a communication between the developed component and the overall EV management system while offering separation of the various research functions.

5.4 Verification and software testing

The importance of software testing lies in ensuring the correct functioning and obtaining appropriate results from the battery health prediction and recommendation functionality.

The verification approach includes the following: data preprocessing, model prediction, recommendation rules, input validation, and communication between the frontend, backend, and machine learning service.

Table 5.2 shows which cases were considered as the primary verification cases for the developed component. These tests offer a systematic approach to assess functionality, input handling, and integration.

Table 6 : Software Testing and Verification Cases

Test ID	**Test Scenario**	**Expected Result**
TC01	Process valid battery degradation data	Data is prepared successfully
TC02	Generate SOH predictions	Predicted SOH values are returned

TC03	Evaluate Random Forest and XGBoost	MAE, RMSE, and R^2 results are generated
TC04	Submit charging and driving information	Relevant recommendations are generated
TC05	Submit invalid or missing input values	Appropriate validation or error response
TC06	Request results through the application	Predictions and recommendations are displayed correctly

For the machine learning models quantitative evaluations are performed based on regression metrics and the functionality of recommendations is verified by evaluating whether the expected recommendation can be made based on the given behavioral conditions.

Integration testing is carried out on the interaction of application services and to validate the output of prediction and recommendations by providing it to the user.

5.5 Deployment and operation

The integrated EV management application will be the host for the developed battery health prediction component.

The prediction service is implemented by transferring the trained machine learning model and can take appropriate input data and giving the estimated battery SOH values. Information on charging and driving is sent separately to the recommendation mechanism.

Requests are routed by the backend to the different services, enabling the user to access the prediction results and recommendations via the application interface.

The machine learning models and recommendation rules can be updated independently of the module deployment structure in case of any changes or improvements.

Operational considerations involve input validation, dealing with appropriate errors, data security and prediction reliability monitoring. There are several possibilities for

future enhancements, such as adding more realistic battery data, more precise forecasting, and optimization for more EV users.

6 RESULTS AND VALIDATION

6.1 Research results

Random Forest and XGBoost regression models were used to test the Adaptive Battery Health Prediction and Personalized Recommendation component. They were both trained and tested independently for forecasting periods of 3 months, 6 months and 12 months.

Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and coefficient of determination (R^2) were used to assess model performance. Table 6.1 shows the results of the independent test data.

Table 7 : Battery Health Prediction Model Comparison Results

Forecasting Period	**Model**	**MAE**	**RMSE**	**R^2**
3 Months	Random Forest	0.017522	0.023360	0.999720
3 Months	XGBoost	0.019656	0.026104	0.999651
6 Months	Random Forest	0.028245	0.039355	0.998999
6 Months	XGBoost	0.030581	0.043160	0.998797
12 Months	Random Forest	0.024652	0.029916	0.999170
12 Months	XGBoost	0.030834	0.036723	0.998749

The results in Table 6.1 indicate that both models performed well in the three forecasting periods. But Random Forest always yielded lower MAE and RMSE values and higher R^2 values than XGBoost.

Random Forest gave an R^2 value of 0.999720 for the 3-month forecast. The numbers of the forecasts for 6 months and 12 months were 0.998999 and 0.999170 respectively.

These results were used to choose Random Forest as the most suitable model for future SOH prediction of the battery.

6.2 Product and technical results

The developed component has 2 main purposes: future battery health prediction and personalized battery maintenance suggestions.

The prediction model provides estimates of SOH for 3, 6, and 12 months. These values are reported via the EV management application, so you can understand what to expect from your battery in the future.

The recommendation mechanism was tested in isolation with predefined charging/driving behavior rules. The frequent fast charging sessions recommended the reduction of unnecessary DC fast-charging sessions, and the aggressive driving pattern suggested smoother accelerating/braking.

The application interface presents the battery health and provides future SOH estimates in an easy-to-use format. The implemented battery health dashboard is as shown in figure 6.1.

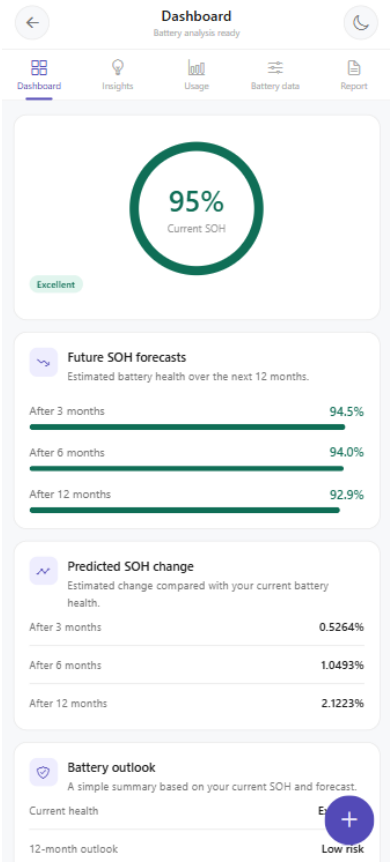


Figure 5 : Battery Health Prediction Dashboard of the EV Management Application

Figure 6.1 is the application interface used to display the current battery SOH, the predicted SOH values for 3, 6 and 12 months, and the estimated change in the battery state. The dashboard displays clearly these results to facilitate the battery management decisions.

6.3 User or stakeholder validation

The developed component will target EV owners to offer easily readable health projections and helpful maintenance guidance.

The application prototype showcases the ability of users to get current and predicted SOH data from one dashboard. Forecasting results are presented, and behavior-based guidance is provided, in ways that will help achieve the usability goals of the proposed component.

Available evaluation does not make a formal user study for this component, however. Thus, additional testing and validation with EV owners and stakeholders are needed to determine the usability, interpretation of predictions and usefulness of recommendations.

6.4 Integration results

The battery health prediction and personal recommendation module were incorporated in the general EV managing application.

The prediction function gives estimated SOH values for 3, 6 and 12 months, and the individual rule-based recommendation function produces battery care advice depending on the user's charging and driving habits.

The developed mobile application interface presents the battery health information in the integrated application. The users can get predictive battery data and other EV management functions.

The integration shows that it's possible to integrate battery health forecasting and personalized recommendations into a single application. To assess deployment reliability and scalability, additional end-to-end testing under actual running conditions would be needed.

6.5 Summary of findings

The experimental findings indicate that the proposed models, Random Forest and XGBoost, can accurately predict the SOH of batteries for the analyzed historical data.

Random Forest outperforms XGBoost during all three forecasting periods with lower MAE and RMSE values, and higher R^2 scores. So, Random Forest was chosen as the ultimate prediction model.

The rule-based recommendation mechanism was also able to generate relevant battery care recommendations for various driving and charging usage patterns. These recommendations are then integrated with future SOH estimates in the integrated EV management application to deliver the developed component.

The results are overall acceptable for the technical viability of the proposed approach. But more testing on representative real-world EV data and direct user validation is required to prove its practical reliability and effectiveness.

7 DISCUSSIONS

7.1 Interpretation of findings

The experimental results show that using historical battery degradation data, the Electric Vehicle (EV) battery State of Health (SOH) can be estimated by Random Forest and XGBoost regression models.

The predictive performance at both models was high for the forecasting periods of 3 months, 6 months and 12 months. On the other hand, Random Forest always achieved lower Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) values than XGBoost. It also obtained higher R^2 values of 0.999720, 0.998999 and 0.999170 for the three forecasting periods respectively.

These results prove that Random Forest was more appropriate for the data set and experimental conditions. Thus, it was selected as the last battery health prediction model.

The rule-based recommendation mechanism was also able to provide meaningful battery maintenance suggestions from user-defined charging and driving patterns. The results validate the potential of future integration of battery health prediction and personalized guidance in an integrated EV management application.

7.2 Comparison with literature baselines and existing solutions

Previous works have shown the effectiveness of machine learning methods in estimating battery health and predicting degradation. The results of the study by Severson et al. [3] showed that battery cycle-life prediction is possible by using a small number of early degradation data points, and the study by Roman et al. [4] proposed a machine learning pipeline for SOH estimation.

Lu et al. [11] delved into deep learning-based battery SOH estimation, and Pandit, Ahlawat [12] compared three machine learning models: Random Forest, XGBoost and Support Vector Machine. They showed how various machine-learning methods can be used for battery health prediction.

Unlike studies mainly related to SOH estimation or cycle-life prediction, the present research aims at evaluating the performance of the Random Forest and XGBoost

models for forecasting periods of three months and at combining the best-performing one with a tailored recommendation approach.

In this study, Random Forest has a better performance in comparison with other studies, as some of the previous one showed that XGBoost outperforms the other one. This difference stresses the importance of the dataset, feature selection and experimental procedures on model performance. Comparisons with previous studies are not possible to make directly due to the differences in the datasets used, the targets of the forecasts and the methods of evaluation.

Existing EV management solutions also provide battery monitoring and performance information. But the proposed component will include multi-horizon SOH forecasting and rules-based recommendations in the context of an EV management application.

7.3 Research contribution

The primary research achievement lies in building and testing regression models using Random Forest and XGBoost algorithms to predict EV battery SOH after 3, 6 and 12 months.

The study offers experimental results, which demonstrate the relative performance of these algorithms based on consistent measures (MAE, RMSE, and R^2). The results align with the choice of the model as the selected historical battery data set was the Random Forest model.

The inclusion of a standalone rule-based recommendation engine is another contribution, while the combination of machine learning-based battery health prediction is integrated. This way, the battery health estimates are correlated with advice for battery maintenance and the existing behavioral information is acknowledged as having its limitations.

The study does not propose any novel machine learning methodology but rather an applied framework of multi-horizon battery health prediction and personalized EV battery management.

7.4 Product contribution

The developed component is a part of the overall Electric Vehicle Management System Using Artificial Intelligence and Machine Learning that offers future battery health estimates and personalized battery care recommendations.

The app displays SOH forecasts in 3, 6 and 12-month time horizon so that users can be aware of the battery health changes that are likely to occur. It is also capable of providing relevant suggestions through charging frequency, fast charging practices and driving behavior.

It is designed as a modular system, which decouples the processing of the predictions, generation of the recommendations and integration with the application. This is a structure that enables maintainability and future enhancements.

The component offers EV owners practical decision-support capabilities, offering battery health forecasts and maintenance recommendations in a single interface.

7.5 Limitations and threats to validity

The experimental results prove that the proposed method has good prediction accuracy, however, there are some limitations that should be noted.

The historically collected battery degradation information was made available to the public for use in the creation of machine learning models. This data set may not be representative of all EV battery chemistries, vehicle types, environmental conditions, and driving patterns.

Moreover, the reliability and confidence in future SOH predictions rely on the quality of the data and the preparation of forecasting targets as well as the assumptions used for representing the 3-month, 6-month and 12-month periods.

R^2 values in the above table are very high, showing good performance on the test data used. But independent validation of the models longitudinally, and due examination of potential data leakage are required to ensure the models are viable to new operating conditions and batteries.

The other drawback is the use of predefined behavioral rules to recommend rather than machine learning driven by relationships between users and battery degradation.

Consequently, the recommendations will only give some practical advice on how to care for the battery but won't specify how much longer the battery will live.

Finally, formal validation with a larger number of EV users is needed to evaluate the long-term usability and effectiveness of the EVs.

7.6 Implications

The results showcase the power of machine learning to enable the future prediction of EV battery health and planning of maintenance tasks.

Predicted SOC values can be helpful to EV owners in identifying potential future battery states and aiding in charging and maintenance decisions. Customized suggestions might also facilitate improved consciousness of battery care techniques.

Technically, the results demonstrated that if the experimental conditions are appropriate, regression methods like Random Forest can yield accurate prediction results. Modular Design also allows for future scalability by integrating with larger EV Management platforms.

The research sets the foundation for future research on larger real-world datasets, better forecasting techniques, and more advanced personalized recommendation techniques. These systems, with further verification, could help with the better management of EV batteries and more sustainable vehicle ownership.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

The research was able to design and test successfully the Adaptive Battery Health Prediction and Personalized Recommendation component for Electric Vehicle Management System (EVM) based on Artificial Intelligence and Machine Learning. The main objective was to predict future EV battery State of Health (SOH) and provide personalized maintenance recommendations to support better battery management.

Two supervised machine learning algorithms, Random Forest and XGBoost Regression were developed and tested for 3-month, 6-month and 12-month battery health prediction. The results of such experiments showed that both models had high predictive accuracy for the selected historical battery data set. Random Forest obtained consistently lower MAE and RMSE, and higher R2 values compared to XGBoost, thus highest prediction accuracy which was preferred.

Furthermore, a rule-based recommendation mechanism was designed to give user-specified charging and driving behavior-based battery care suggestions. The prediction and recommendation features have been embedded in the complete EV management application to display relevant battery health data in a single application.

In conclusion, the study highlights the feasibility of integrating machine learning for SOH prediction with individual battery maintenance recommendations. Unfortunately, future testing of the model in actual conditions is needed for validation of the reliability of long-term predictions and the usefulness of the recommendations.

8.2 Achievement of objectives

The research addressed the main objectives established at the beginning of the study.

The first goal was to analyze and create historical battery degradation data for machine learning. Data preprocessing, feature preparation, and formulating of appropriate SOH prediction inputs were used to overcome this.

The second goal was to create multi-horizon battery health prediction models. Three regressions models were built and tested: Random Forest and XGBoost with 3-, 6- and 12-month forecasting periods.

The third was to evaluate the accuracy of model predictions using MAE, RMSE and R^2. Results of the experiments revealed that Random Forest is better than XGBoost in all three forecasting periods.

The fourth goal was to offer customized battery care suggestions. A charging rule-based mechanism was designed to analyze the charging and driving behavior and offer battery care suggestions based on the information.

The fifth goal was to embed the designed component in the entire EV management application. This has been accomplished at the prototype level by providing battery health forecasts and personalized recommendations through the application interface.

The goal was to assess the performance of components and to understand any limitations. There was evidence that the models were technically feasible, but there were some limitations concerning the representativeness of the datasets, assumptions of the models, and validation in the real world.

Thus, the overall technical goals of the research were met, but more user validation and field testing would be required, in a broader sense.

8.3 Individual research and product contributions

The primary individual research contribution comes in the development and comparative assessment of Random Forest and XGBoost regression methods for predicting the future SOH of EV batteries. The study offers experimental proof of the fact that Random Forest is more appropriate algorithm for the selected historical data of battery degradation.

Another research contribution is the fusion of a multi-horizon battery health forecasting with an independent rule-based recommendation mechanism. This approach is useful for dealing with the reality that maintenance advice to batteries must be given as well as numerical health predictions and assumes no a priori knowledge of whether behavioral variables were used in the model training set.

Product contribution refers to the integration of the battery health prediction and personalized recommendation feature into the EV management application. The

component allows the user to see displayed the estimated future SOH values for three, six and 12 months, as well as the appropriate charging and driving suggestions.

The modular design also facilitates maintainability and the future development of the prediction models and the recommendation mechanism. All these efforts create a solid basis for intelligent EV battery health management.

8.4 Future directions

The experimental results of the developed component were encouraging but there are several improvements which can be thought of in future research.

First, a much larger and more diverse real-world EV battery dataset could be leveraged to train and evaluate the prediction models. To assess generalizability of models, different battery chemistries, vehicle models, operating temperatures, and usage conditions should be included.

Second, more work is needed on longitudinal datasets of batteries that provide reliable time information to confirm the accuracy of the 3-, 6-, and 12-month SOH forecasts. More testing on independent batteries would also be useful in determining the accuracy of the predictions under various operating conditions.

Third, the personalized recommendation system could be further enhanced by taking into consideration real charging and driving data from EV users. Machine-learning-based personalization can be pursued by using the appropriate data to offer more personalized recommendations.

Additional enhancements could include live battery status tracking, ongoing model performance assessment, better data presentation and a secure connection with vehicle telemetry systems.

Lastly, there would be significant user testing and deployment studies to assess system usability, reliability and the usefulness of the battery health predictions and maintenance recommendations produced. The enhancements would contribute to the creation of a more trustworthy and adaptable EV battery management solution.

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GLOSSARY

Battery Degradation - Deterioration in a battery's capacity or performance over time, from aging, charging cycles, temperature, and operating conditions.

Battery Health Prediction - This is the process of predicting what a battery will be like in the future based on past degradation data and predictive models.

Calendar Age - Battery degradation occurs because of age, even if it is not in use.

Charging Behavior-How often, how long, and how much fast charging an EV user is using.

Rising Badger Count - An increase in the number of badgers.

Data Preprocessing-This is the process of cleaning and preparing raw data by dealing with missing values, removing invalid records, etc., and transforming features for machine learning.

Driving Behavior-The driving habits of an EV user, such as accelerating and braking, driving speed, and distance traveled over the course of a day.

Ensemble Learning-It is a machine learning method used to enhance the performance of the prediction models by using multiple models.

Feature Engineering - Selection, creation, or transformation of relevant input variables to enhance the performance of machine learning models.

Forecasting Horizon - The time frame that a prediction is made for, e.g., 3, 6, or 12 months.

Gradient Boosting - An ML method that sequentially adds models, with each model trying to minimize the error of the previous model.

Model Evaluation - The evaluation of the performance of a machine learning prediction by applying the right statistical measure.

Personalized Recommendation - Battery maintenance recommendations based on personal driving and charging habits.

Random Forest Regression - An ensemble machine learning algorithm that can estimate continuous numerical values using multiple decision trees.

Regression Model - An ML model that can predict continuous values like battery health percentages.

Rule-Based Recommendation Engine - A system that uses predefined conditions and rules to make relevant recommendations of user data.

State of Health Estimation - The process of estimating the state of a battery based on its capacity when compared to the original battery capacity.

Training Dataset - A set of data that is used to teach machine learning models about the relationships between features in the dataset and the target values.

Testing Dataset - A separate subset of data that is used to examine the performance of a trained model on unseen data.

XGBoost Regression - A gradient boosting algorithm that sequentially trains decision trees to help improve the prediction of numerical data.